Mechanism Design: the Revelation Principle

Mechanisms and Incentives

A large part of last century's economic theory can be seen as the search for a mathematical restatement of the Adam Smith's metaphor, the idea that market forces are the artifact of some invisible hand

Market prices are the fundamental force conveying all relevant information and driving individual behaviors

The word "incentives" does NOT appear in Schumpeter's History of Economic Analysis (1954)

Incentive Theory (1)

Broadly speaking, “Incentive Theory” refers to all those situations in which the allocation of economic resources is **not** driven by (market) prices

The theory was born with a normative aim: how to “design” economic institutions in such a way that they induce agents to behave in accordance with some pre-specified objective?

1. Economic calculations under socialism (Hurwicz (1973))
2. Optimal Taxation (Mirrlees (1971))
3. Internal organization of firms (Holmstrom (1979), Williamson (1985))

Incentive Theory (2)

In the last decades, the methods and insights of mechanism design have been applied from a positive point of view:

1. Extending General Equilibrium Theory to cope with asymmetric information
2. Financial and labor markets under private information (Akerlof (1970), Rothschild-Stiglitz (1976))
3. Limited commitment to future obligations (Kehoe-Levine (1993))

Mechanism Design

1. Correlated Equilibria

2. A generalized Principal-Agent(s) model: Myerson (1982)

3. A Canonical Application: Bilateral Trade

4. The Taxation Principle: Rochet (1985)

We start with finite strategic games

$$G = \{\mathcal{N}, (A_i)_{i\in\mathcal{N}}, (U_i)_{i\in\mathcal{N}}\}$$

with $\mathcal{N} = \{1, \cdots, N\}$, $U_i : A \to \mathbb{R}$

The Notion of "Device"

Example 1 The *Battle of Sexes (BoS)*

P1 \ *P2*	*Bach*	*Stravinsky*
Bach	$2, 1$	$0, 0$
Stravinsky	$0, 0$	$1, 2$

The game admits a completely mixed NE equilibrium $\{(2/3, 1/3), (1/3, 2/3)\}$ *with corresponding payoffs* $(2/3, 2/3)$.

Interpretation: Each player's (pure) action is chosen according to a **private** *and* **independent** *random signal*

The "extended" game in which P1 receives a signal in $S^1 = \{B, S\}$ with probabilities $(2/3, 1/3)$ and P2 receives a signal in $S^2 = \{B, S\}$ with probabilities $(1/3, 2/3)$ admits a **pure strategy** NE which reproduces the allocation of the above mixed equilibrium.

Can one reproduce additional equilibrium outcomes by considering alternative devices?

Public Correlating Devices

Consider now a perfectly correlated signal in $S^1 = S^2 = \{x, y\}$ with probabilities $(1/2, 1/2)$. The strategies

- Play Bach upon receiving $\{x\}$
- Play Stravinsky upon receiving $\{y\}$

implement a new (**efficient**) equilibrium allocation, yielding expected payoffs $(3/2, 3/2)$.

These payoffs belong to the convex hull of the NE of G.

Example 2: The Chicken Game (Aumann)

A Movie: "Rebel without a Cause"

$P1 \setminus P2$	C	D
C	$6, 6$	$2, 7$
D	$7, 2$	$0, 0$

Chicken $\equiv$ "Always Swerve", Daring $\equiv$ "Never Swerve"

The game admits a completely mixed NE equilibrium $\{(2/3, 1/3), (2/3, 1/3)\}$ with corresponding payoffs $(14/3, 14/3)$.

A General Device

A random signal in $\Omega = \{L, M, H\}$ with probability $1/3$ each.

- P^i is a partition of Ω for player i

Partition $\equiv$ Countable collection of disjoint sets which union is Ω

- $p^i(\omega)$ is an element of P^i containing $\omega \in \Omega$

The Augmented Game

Consider the following extenive-form game:

1. $\omega \in \Omega$ is chosen according to the above distribution
2. Each player $i \in \mathcal{N}$ is (privately) informed about $p^i(\omega)$
3. G is played and its payoffs are implemented

A Correlated Equilibrium of G

Consider the partitions P^1 and P^2 such that

- $p^1(H) = \{H\}$, $p^1(L) = p^1(M) = \{L, M\}$
- $p^2(L) = \{L\}$, $p^1(H) = p^2(M) = \{H, M\}$

Claim: The strategies

- Play $\{D\}$ if H and $\{C\}$ otherwise for P1
- Play $\{D\}$ if L and $\{C\}$ otherwise for P2

constitute a (pure strategy) NE of G

The NE induces a probability distribution over actions

$P1 \setminus P2$	C	D
C	$1/3$	$1/3$
D	$1/3$	0

The corresponding payoffs $(5, 5)$ do NOT belong to the convex hull of the set of NE of G

Correlated Equilibrium: Definition

Consider the finite strategic game $G = \left\{N, (A_i)_{i\in\mathcal{N}}, (U_i)_{i\in\mathcal{N}}\right\}$

A correlating device is the vector $\gamma = \{\Omega, q, (P^i)_{i\in\mathcal{N}}\}$, where q is the probability distribution over Ω.

The device induces the following (extensive form) game G^γ

1. ω is chosen from Ω according to q
2. Each player $i \in \mathcal{N}$ is (privately) informed about $p^i(\omega)$
3. G is played and its payoffs are implemented

Correlated Equilibrium: Definition

Strategy in G^γ: $\sigma_i : \Omega \to A_i$, P^i-measurable: $\sigma_i(\omega) = \sigma_i(\omega')$ if $\omega' \in p^i(\omega)$

Each player i behaves given his private information on the random signal which is selected before the beginning of G

Aumann (1974): A correlated equilibrium of G is the pair (γ, σ) in which σ is an equilibrium of the game G^γ. That is:

$$\sum_{\omega\in\Omega} q(\omega) U_i(\sigma_i(\omega), \sigma_{-i}(\omega)) \geq \sum_{\omega\in\Omega} q(\omega) U_i(\tau_i(\omega), \sigma_{-i}(\omega)) \tag{1}$$

for each $i \in \mathcal{N}$ and for each $\tau_i \in A_i$

EU $\Longrightarrow$ this is equivalent to consider optimality for each ω

Correlated Equilibrium: Results (1)

Proposition 1 (Correlated $\rightarrow$ Nash)

For each (mixed strategy) NE $\alpha \in \Delta A$ of the game G, there is a correlated equilibrium (γ, σ) such that the distribution on A induced by σ is exactly α

Proof: Consider the following (γ, σ)

1. $\Omega = A$; and $q = \underset{i}{\times} \alpha_i(a_i)$
2. Each P^i is generated by A_i. For each i and for each $b_i \in A_i$:
   - $p^i(b_i) = \{a \in A : a_i = b_i\}$
   - P^i is the collection of all $|A_i|$ sets $p^i(b_i)$
3. $\sigma_i(a) = a_i$

Correlated Equilibrium: Results (2)

Proposition 2 (Convex Combination of Correlated)

Any convex combination of correlated equilibrium payoffs of G is a correlated equilibrium payoff of G

Proposition 2: Remarks

1. CE as a device to “convexify” the set of NE (Battle of Sexes)

2. There exist CE which do not belong to the convex hull of NE (Chicken Game)

Correlated Equilibrium: Results (3)

Proposition 3 (Canonical Game)

Any distribution over outcomes that obtains in a correlated equilibrium of G also obtains in a correlated equilibrium in which players follow their "recommended" actions

Proof: Let (γ, σ) be a correlated equilibrium of G.
Consider the pair $(\gamma' = \{\Omega', q', (P'^i)_{i \in \mathcal{N}}\}, \sigma')$:

1. $\Omega' = A$, and $q'(a) = q(\omega \in \Omega : \sigma(\omega) = a)$
2. $\sigma'_i(a) = a_i$

Correlated Equilibrium: Canonical Representation

Following Aumann (1974), we denote (γ', σ') a *canonical correlated equilibrium can be interpreted as*

1. *A mediator selects $\{a\}$ according to q and privately recommends $\{a_i\}$ to each player i*
2. *Players are not forced to obey the mediator, but $\{a\}$ is selected in such a way that player i cannot profitably deviate from the recommendation, i.e. $\hat{a}_i = a_i$ maximizes $U_i(\hat{a}_i, a_{-i})$.*

Example: the distribution over actions in the Chicken Game

A generalized Principal-Agent(s) model: Myerson (1982)

- One principal and a set $\mathcal{N} = \{1, 2, \ldots, N\}$ of agents
- Each agent i has access to **private information** (encapsulated by his type, $\theta_i \in \Theta_i$) and can perform a **private action** ($a_i \in A_i$)
- The (partial) control of the principal over the final outcome is represented by the decision set D **which may include agents' observable (contractible) actions**

Bayesian Incentive Problem

A *Bayesian incentive problem* (BIP) is the array $\gamma = (\mathcal{N} \cup \{0\}, A, , p(\theta), U)$, where

- $\mathcal{N} = \{1, 2, \ldots, N\}$ is the set of agents, and we denote by $i = 0$ the principal;
- $A = A_1 \times \ldots \times A_N$ is the set of agents' actions;
- $D \times A$ is the set of allocations;
- $\Theta = \Theta_1 \times \ldots \times \Theta_N$ is the set of agents' types.

Bayesian Incentive Problem (cont'd)

– $p(\theta)$ is the common prior, i.e. a commonly known probability distribution over the types that each player uses to compute his/her beliefs about other players' types;

– $U = (U_0, U_1, \ldots, U_N)$ where $U_i : D \times A \times \Theta \to \mathbb{R}$ for $i = 0, 1, \ldots, N$ is the vector of players' payoffs.

Coordination Mechanism

The principal designs a *coordination mechanism* $\gamma = ((R, M), q)$, where

- $R = R_1 \times \ldots \times R_N$ is the set of signals (recommendations) he sends to agents (**analogy with "states"**)
- $M = M_1 \times \ldots \times M_N$ is the set of reports (messages) that he receives from agents
- $q : D \times M \times R \to \mathbb{R}$ is a probability function, assigning a probability to each combination of messages and principal's decisions, conditional on the reports received from the agents.

Applications

This general setting covers many fundamental applications:

1. $\underline{N = 1}$: This corresponds to the class of "Principal-Agent" models (Laffont-Martimort (2002))
2. $\underline{N > 1 \text{ and } A = \{a\}}$ "pure" **incomplete** information: Auction Theory (Vickrey(1961), Myerson (1981), Cremer-McLean (1985))
3. $\underline{N > 1 \text{ and } \Theta = \{\theta\}}$ -"pure" **complete** information: Teams and Parnterships (Holmstrom (1982))

The Continuation Game

The mechanism γ is publicly observed by agents. Each $\gamma \in \Gamma$ induces an incomplete information (continuation) game amongst agents.
Agents' strategies can be represented as:

- $\rho_i : \Theta_i \to \Delta(M_i)$
- $\sigma_i : R_i \times \Theta_i \to \Delta(A_i)$

The Continuation Game: Payoffs

The expected utility to agent i given the strategies $(\rho, \sigma) = ((\rho_{-i}, \sigma_{-i}), (\rho_i, \sigma_i))$, after he learns his type (interim), is:

$$\sum_{\theta_{-i} \in \Theta_{-i}} \sum_{d \in D} \sum_{r \in R} p(\theta_{-i}|\theta_i) q(d, r|\rho(\theta)) U_i((d, \sigma(r, \theta)), \theta) \qquad (2)$$

where $p(\theta_{-i}|\theta_i)$ denotes the probability that agent i attributes to the event that the vector of types $\theta_{-i} = (\theta_1, \ldots, \theta_{i-1}, \theta_{i+1}, \ldots, \theta_N)$ is observed, given that his own type is θ_i.

Direct Mechanisms

The number of possible mechanisms is generally large, and difficult to characterize, since the sets R and M may have a very complex structure.

A simple class of mechanisms:

Definition 1 (Direct Mechanism)

A mechanism $\gamma \in \Gamma$ *is* **direct** *if* $M_i = \Theta_i$ *and* $R_i = A_i$ *for all* i.

Including agents' participation

Denote $\{Y, N\}$ the decision of agent i to participate with the principal. The decision is payoff relevant, hence we write:

$$U_i : \{Y, N\}^N \times A \times \Theta \to \mathbb{R}$$

for $i = 0, 1, \ldots, N$

The principal perfectly observes the set of agents who participate with him $\Longrightarrow$

$$\hat{D} : \{Y, N\}^N \to D$$

Participation and Communication

Each M_i set contains the $\emptyset$ element representing "Agent i does not participate with the principal"

Most applications rely on the following

Assumption P: The set of participation and communication decisions for each agent i is such that $m_i = 0$ *iff* agent i does not participate with the principal

In the sequel we ignore participation unless otherwise specified

Incentive-Compatible Direct Mechanisms

In a direct mechanism, each agent i must report a type $\theta_i \in \Theta_i$ to the principal, and, in return, the principal will send each agent a suggested decision $a_i \in A_i$.
An additional restriction:

Definition 2 (Incentive Compatibility)

In a direct mechanism q, agent i is **honest** *and* **obedient** *if (ρ_i^*, σ_i^*) are such that*

$$\rho_i^*(\theta_i) = \theta_i, \quad \text{and} \quad \sigma_i^*(a_i, \theta_i) = a_i$$

for all $\theta_i \in \Theta_i$ and $a_i \in A_i$ and for all $i \in \mathcal{N}$.

Incentive-Compatible Mechanisms (cont'd)

Definition 3 (Incentive-Compatible Mechanism)

A direct mechanism is **Bayesian incentive-compatible** *if the honest-obedient participation strategies* $(\rho_i^*, \sigma_i^*)_{i \in \mathcal{N}}$ *form an equilibrium of the continuation game.*

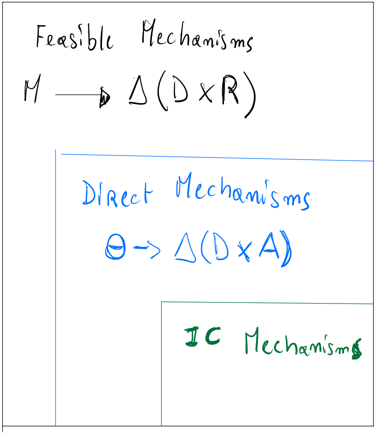
Incentive-Compatible Mechanisms (cont'd)

Each type of each agent does not gain by deviating from his honest and obedient behavior :

$$\begin{aligned}\mathbb{E}[U_i(d,a,\theta)|\theta_i] &= \sum_{\theta_{-i}\in\Theta_{-i}} \sum_{d\in D} \sum_{a\in A} p(\theta_{-i}|\theta_i) q(d,a|\theta) U_i(d,a,\theta) \\ &\geq \sum_{\theta_{-i}\in\Theta_{-i}} \sum_{d\in D} \sum_{a\in A} p(\theta_{-i}|\theta_i) q(d,a|\theta_{-i},\hat{\tau}_i) U_i((d,a_{-i},\hat{\sigma}_i(a_i)),\end{aligned} \tag{3}$$

for all $i \in \mathcal{N}$, $\theta_i \in \Theta_i$, $\hat{\tau}_i \in \Theta_i$ and $\hat{\sigma}_i : A_i \to A_i$.

Feasible Mechanisms



The Principal's Problem

If we focus (w.l.o.g., as we will show) on incentive-compatible direct mechanisms, the principal's problem can be expressed as a constrained maximization problem of the form

$$\max_{q} \ \mathbb{E}[U_0(d,a,\theta)] = \sum_{\theta\in\Theta}\sum_{d\in D}\sum_{a\in A} p(\theta)q(d,a|\theta)U_0(d,a,\theta)$$

$$\text{s.t.} \quad (3)$$

where the objective function is the expected utility of the principal, given that he adopts mechanism q (which must be a valid conditional probability distribution).

This is a linear programming problem.

The Revelation Principle

Proposition 4 (*Revelation Principle*)

Given any equilibrium of the agents' continuation game $(\rho_i, \sigma_i)_{i=1}^{n}$ induced by any mechanism $((R_i, M_i)_{i=1}^{N}, q)$, there exists an incentive-compatible direct mechanism q^* in which the principal gets the same expected utility (when the agents are honest and obedient) as in the given equilibrium of the given mechanism.

Thus, the optimal incentive-compatible direct coordination mechanism is also optimal in the class of all coordination mechanisms.

Proof of Proposition 4

Since q is fixed throughout the proof, we omit the dependence of the strategies on the mechanism.

Given q, $\sigma_i(r_i, \theta_i)$ identifies the (probability distribution over) action(s) that agent i picks based on his type θ_i and the signal r_i received from the principal.

$\Rightarrow$ The original mechanism $q(d, r|\rho(\theta))$ effectively induces a probability distribution over the agents' actions a_i (with $i = 1, \ldots, n$) through the measurable strategy functions $(\sigma_i)_{i \in \mathcal{N}}$.

We exploit this connection to recover the probability distribution over $D \times A$

Proof of Proposition 4 (cont'd)

Compute the probability that, under the original mechanism, the equilibrium actions a are observed, given the agents' messages $\rho(\theta)$:

$$\begin{aligned}\mathbb{P}(d, a|\rho(\theta)) &= \mathbb{P}(d, \{\sigma(r, \theta) = a\}|\rho(\theta)) \\ &= \mathbb{P}(d, \{r \in \sigma^{-1}(a, \theta)\}|\rho(\theta)) \\ &= \sum_{r \in \sigma^{-1}(a,\theta)} q(d, r|\rho(\theta))\end{aligned}$$

where $\sigma^{-1}(a, \theta)$ is the set of (vectors of) signals that induce the (vector of) actions a through the equilibrium strategy $\sigma(r, \theta)$, and the mechanism q. More formally,

$$\sigma^{-1}(a, \theta) = \{r \mid \sigma_i(r_i, \theta_i) = a_i,\ \forall i = 1, \ldots, N\}$$

Proof of Proposition 4 (cont'd)

Define the new **direct** mechanism $q^* : \Theta \times D \times A \to \mathbb{R}$

$$q^*(d, a|\theta) \equiv \mathbb{P}(d, a|\rho(\theta))$$

That is:

- The principal chooses d and agents choose a when $\theta \in \Theta$ realizes
- The principal's decisions and signals are determined by messages ρ through q
- Each agent uses σ_i to translate signals into actions

Proof of Proposition 4 (cont'd)

By construction, $q^* : D \times A \times \Theta \to \mathbb{R}$ is the probability measure, derived from q, that attributes to each decision vector in $D \times A$ the same probability as the original mechanism, given that the individuals report type θ.

$\Rightarrow$ If we let players randomizing directly over decisions $(d, a) \in D \times A$ and using as measure q^*, we get the **same (random) allocation** as in the original mechanism, where we used the measure q, but randomized over the principal's signals and decisions $(r, d) \in R \times D$.

Proof of Proposition 4 (cont'd)

Summing up...

1. We start from the messages that each agent i would have sent to the principal under the original mechanism, $\rho_i(\theta_i)$
2. We compute, given ρ, the signals that the principal would have sent to the agents under the old mechanism to induce the profile (d, a)
3. Finally, we construct the new mechanism

$$q^*(d, a|\theta) \equiv \mathbb{P}(d, a|\rho(\theta))$$

Proof of Proposition 4 (cont'd)

Intuition

We have to prove that this new mechanism is incentive compatible, i.e. that it induces the agents to report their type honestly and behave according to the principal's recommendation.

Intuitively,

- The agents do not have to lie, because the mechanism "lies" for them;
- The agents do not have to compute their optimal response to the principal suggestion, because the mechanism directly suggests it to them.

Proof of Proposition 4 (cont'd)

Formal argument

We prove that q^* satisfies Condition 3 by contradiction.

Suppose an agent i of type τ_i has an incentive to **lie** (declare type $\hat{\tau}_i \in \Theta_i$) and/or to **disobey** (choose $\hat{\sigma}(a_i) \in A_i$, when recommended a_i):

$$\sum_{\theta_{-i} \in \Theta_{-i}} \sum_{d \in D} \sum_{a \in A} p(\theta_{-i}|\tau_i) q^*(d, a|\theta) U_i(d, a, \theta)$$
$$< \sum_{\theta_{-i} \in \Theta_{-i}} \sum_{d \in D} \sum_{a \in A} p(\theta_{-i}|\tau_i) q^*(d, a|\theta_{-i}, \hat{\tau}_i) U_i((a_{-i}, \hat{\sigma}_i(a_i)), \theta)$$

where $\theta = (\theta_{-i}, \tau_i)$ and $\hat{\sigma}_i : A_i \to A_i$.

Proof of Proposition 4 (cont'd)

Formal argument

In terms of the original mechanism, q, these strategies correspond to

$$\tilde{\rho}_i(\theta_i) \equiv \begin{cases} \rho_i(\hat{\tau}_i) & \text{if } \theta_i = \tau_i \\ \rho_i(\theta_i) & \text{otherwise} \end{cases}$$

and

$$\tilde{\sigma}_i(r_i, \theta_i) \equiv \begin{cases} \hat{\sigma}_i(r_i, \hat{\tau}_i; \theta_i) & \text{if } \theta_i = \tau_i \\ \sigma_i(r_i, \theta_i) & \text{otherwise} \end{cases}$$

Proof of Proposition 4 (cont'd)

Formal argument

Then we can write:

$$\sum_{\theta_{-i}} \sum_{d \in D} \sum_{r \in R} p(\theta_{-i}|\tau_i) q(d, r|\rho(\theta)) U_i((d, \sigma(r, \theta)), \theta)$$

$$= \sum_{\theta_{-i}} \sum_{d \in D} \sum_{a \in A} p(\theta_{-i}|\tau_i) q^*(d, a|\theta) U_i(d, a, \theta)$$

$$< \sum_{\theta_{-i}} \sum_{d \in D} \sum_{a \in A} p(\theta_{-i}|\tau_i) q^*(d, a|\theta_{-i}, \hat{\tau}_i) U_i((d, a_{-i}, \hat{\sigma}_i(a_i)), \theta)$$

$$= \sum_{\theta_{-i}} \sum_{d \in D} \sum_{r \in R} p(\theta_{-i}|\tau_i) q(d, r|\rho_{-i}(\theta_{-i}), \tilde{\rho}_i(\tau_i)) U_i((d, \sigma_{-i}(r_{-i}, \theta_{-i}), \tilde{\sigma}_i(r_i, \tau_i)), \theta)$$

This contradicts the assumption that $(\rho_i, \sigma_i)_{i=1}^N$ is an equilibrium of the original mechanism q.

Remark 1

Consider the special case $D = \{d\}$ and $\Theta = \{\theta\}$

The principal does not control any decision and the agent has no private information

$\Longrightarrow$ The incentive-compatible mechanisms and the agents' continuation equilibria are canonical Correlated Equilibria (Aumann, 1974)

Remark 2: Implementation Theory

Our generalized principal-agent problem may be viewed as a hybrid between a cooperative and a non-cooperative game. The agents are assumed to act non-cooperatively as utility maximizers who will passively accept any Nash equilibrium of any game which the principal chooses. The possibilities for communication and cooperation are assumedto be entirely controlled by the principal. Thus, when we identify a particular individual as the 'principal', we mean that we want to know what would be the best possible cooperative arrangement for the purposes of this individual, if he had all of the bargaining ability

Myerson (1982)

Remark 2: Implementation Theory

The principal faces **three** privately informed agents, and $D = \{a, b, c\}$.

$\Theta_1 = \{\theta\}, \Theta_2 = \Theta_3 = \{\theta_L, \theta_H\}$, and $pr(\theta_L, \theta_L) = pr(\theta_H, \theta_H) = 1/2$

Claim: there exist truthful equilibrium allocations in the direct mechanism game G^D that cannot be supported in all message games

Remark 2: Payoffs

Payoffs in (θ_L, θ_L)

- $U_0(a, \theta_L, \theta_L) = U_1(a, \theta_L, \theta_L) = 0$; $U_2(a, \theta_L, \theta_L) = U_3(a, \theta_L, \theta_L) = 4$
- $U_0(b, \theta_L, \theta_L) = U_1(b, \theta_L, \theta_L) = 4$; $U_2(b, \theta_L, \theta_L) = U_3(b, \theta_L, \theta_L) = 3$
- $U_0(c, \theta_L, \theta_L) = U_1(c, \theta_L, \theta_L) = 3$; $U_2(c, \theta_L, \theta_L) = U_3(c, \theta_L, \theta_L) = 0$

Payoffs in (θ_H, θ_H)

- $U_0(a, \theta_H, \theta_H) = U_1(a, \theta_H, \theta_H) = 4$; $U_2(a, \theta_H, \theta_H) = U_3(a, \theta_H, \theta_H) = 3$
- $U_0(b, \theta_H, \theta_H) = U_1(b, \theta_H, \theta_H) = 0$; $U_2(b, \theta_H, \theta_H) = U_3(b, \theta_H, \theta_H) = 4$
- $U_0(c, \theta_H, \theta_H) = U_1(c, \theta_H, \theta_H) = 3$; $U_2(c, \theta_H, \theta_H) = U_3(c, \theta_H, \theta_H) = 0$

Remark 2: The G^D Game

Lemma: There exists an equilibrium of G^D implementing $\{a\}$ in (θ_L, θ_L) and $\{b\}$ in (θ_H, θ_H): P gets his minimal payoff

Proof: The message space of each agent i is Ω^i, for $i = 1, 2, 3$. Consider the mechanism:

$$\gamma(\hat{\theta}_2, \hat{\theta}_3) = \begin{cases} a & \text{if} \quad (\hat{\theta}_2, \hat{\theta}_3) = (\theta_L, \theta_L) \\ c & \text{if} \quad (\hat{\theta}_2, \hat{\theta}_3) \in \{(\theta_L, \theta_H), (\theta_H, \theta_L)\} \\ b & \text{if} \quad (\hat{\theta}_2, \hat{\theta}_3) = (\theta_H, \theta_H) \end{cases}$$

The agents' game also has an untruthful equilibrium, in which $U_0 = U_1 = 4$

Remark 2: Enlarging Message Spaces

Consider a game in which $M_1 = \{left, right\}$, and $M_i = \Theta_i$ for $i = 2, 3$
The mechanism:

$$\gamma(left, \hat{\theta}_2, \hat{\theta}_3) = \begin{cases} b & \text{if} \quad (\hat{\theta}_2, \hat{\theta}_3) = (\theta_L, \theta_L) \\ c & \text{if} \quad (\hat{\theta}_2, \hat{\theta}_3) \in \{(\theta_L, \theta_H), (\theta_H, \theta_L)\} \\ a & \text{if} \quad (\hat{\theta}_2, \hat{\theta}_3) = (\theta_H, \theta_H) \end{cases}$$

and

$$\gamma(right, \hat{\theta}_2, \hat{\theta}_3) = \begin{cases} b & \text{if} \quad (\hat{\theta}_2, \hat{\theta}_3) = (\theta_L, \theta_H) \\ c & \text{if} \quad (\hat{\theta}_2, \hat{\theta}_3) \in \{(\theta_L, \theta_L), (\theta_H, \theta_H)\} \\ a & \text{if} \quad (\hat{\theta}_2, \hat{\theta}_3) = (\theta_H, \theta_L) \end{cases}$$

Remark 2: Non-Robustness

One can check that all equilibria of the agents' message game implement $\{b\}$ **in** (θ_L, θ_L) **and** $\{a\}$ **in** (θ_H, θ_H)

At equilibrium, A1 send the message $\{left\}$, and uses the message $\{right\}$ to kill "undesirable" outcomes <u>on the equilibrium path</u>

The indirect mechanism <u>uniquely</u> implements the optimal allocation for P

Remark 3: The Need for Random Mechanisms

$D = \{A, B, C\}$, $N = 2$, but only A1 is privately informed: $\Theta_1 = \{\theta_L, \theta_H\}$

P's Payoffs

- $U_0(A, \theta_L) = 1,\ U_0(B, \theta_L) = U_0(C, \theta_L) = 0$
- $U_0(A, \theta_H) = 0,\ U_0(B, \theta_H) = 1,\ U_0(C, \theta_H) = 0$

A1's Payoffs

- $U_1(A, \theta_L) = 1,\ U_1(B, \theta_L) = U_1(C, \theta_L) = 0$
- $U_1(A, \theta_H) = 0,\ U_1(B, \theta_H) = 1,\ U_1(C, \theta_H) = 1$

A2's Payoffs

- $U_2(A, \theta_L) = 0,\ U_2(B, \theta_L) = -1,\ U_2(C, \theta_L) = 1$
- $U_2(A, \theta_H) = 0,\ U_2(B, \theta_H) = -1,\ U_2(C, \theta_H) = 1$

Remark 3: Enlarging Message Spaces AGAIN

Consider a game in which $M^1 = \{m_1, m_2, m_3\}$, $M^2 = \{\emptyset\}$ and the mechanism:

$$\gamma(m) = \begin{cases} A & \text{if} \quad m = m_1 \\ B & \text{if} \quad m = m_2 \\ C & \text{if} \quad m = m_3 \end{cases}$$

It induces an equilibrium in which:

1. Type θ_L of A1 sends m_1 with probability one
2. Type θ_H of A1 sends m_2 and m_3 with probability $1/2$ each

Remark 3: Payoffs

1. P gets 3/4
2. θ_L and θ_H of A1 get 1
3. A2 gets zero

It is straightforward to see that if the principal uses a <u>deterministic</u> direct mechanism $\hat{\gamma}$ at least one player is worse off: to guarantee θ_L his payoff of one, we need $\hat{\gamma}(\theta_L) = A$, but for any $\hat{\gamma}(\theta_H) \in \{A, B, C\}$ either P or A2 loses.

A Canonical Application: Bilateral Trade

We consider the trade of a single indivisible good between **two** players: a Seller (S) and a Buyer (B).

Trades may be limited by private information: $\Theta_B = \Theta_S = \{L, H\}$. Types are independently distributed, and $prob(\theta_B = H) = b_H$, $prob(\theta_S = L) = s_L$

Preferences are:

$$U_B(\theta) = v_{\theta_B} - P$$
$$U_S(\theta) = P - c_{\theta_S}$$

with $v_H > v_L > 0$ and $c_H > c_L$

A Benchmark: $\Theta_S = \{\theta\}$ and $c_\theta = 0$

The seller is uninformed of the buyer's valuation for the good.

A mechanism is a (deterministic) take-or-leave-it price $P \in \mathbb{R}_+$

<u>The mechanism which maximizes the seller's expected payoff $\tilde{U}^S$:</u>

1. $P = v_L \implies$ both types of B accept $\implies \tilde{U}^S = v_L$
2. $P = v_H \implies$ only type H of B accepts $\implies \tilde{U}^S = b_H v_H$

If $b_H v_H > v_L$, optimality for the seller conflicts with (ex-post) efficiency, which requires both types of the buyer trading the object.

One-sided Private Information (cont'd)

The mechanism which maximizes each buyer's payoff:

This is trivially $P = 0$: no B's type is excluded.

$\Longrightarrow$ **A conflict between allocative efficiency and the distribution of informational rents.**

Naive interpretation: if all bargaining power of the uninformed party could be removed, ex-post efficiency would obtain.

The General Case: Two-sided Private Information

We consider a particular case:

$$v_H > c_H > v_L > c_L$$

That is: efficiency requires that trade always occurs unless $\underline{v_L < c_H}$

A mechanism is $P(\theta^B, \theta^S) : \{L, H\}^2 \to \mathbb{R}^2_+$

We focus on direct mechanisms inducing efficient trading $\Longrightarrow$ no trade if (L, H) is reported and P_{LH} is implemented

Two-sided Private Information: Ex-Ante Participation

The Seller:

$$
\begin{aligned}
&(IC_L) \quad b_H[P_{HL} - c_L] + (1 - b_H)[P_{LL} - c_L] \geq b_H[P_{HH} - c_L] + (1 - b_H)P_{LH} \\
&(IC_H) \quad b_H[P_{HH} - c_H] + (1 - b_H)P_{LH} \geq b_H[P_{HL} - c_H] + (1 - b_H)[P_{LL} - c_H] \\
&(IR) \quad (1 - s_L)\big[b_H[P_{HH} - c_H] + (1 - b_H)P_{LH}\big] + s_L\big[b_H[P_{HL} - c_L] + (1 - b_H)P_{LL} - c_L\big] \geq 0
\end{aligned}
$$

The Buyer:

$$
\begin{aligned}
&(IC_L) \quad s_L[v_L - P_{LL}] - (1 - s_L)P_{LH} \geq s_L[v_L - P_{HL}] + (1 - s_L)[v_L - P_{HH}] \\
&(IC_H) \quad s_L[v_H - P_{HL}] + (1 - s_L)[v_H - P_{HH}] \geq s_L[v_H - P_{LL}] - (1 - s_L)P_{LH} \\
&(IR) \quad b_H\big[s_L[v_H - P_{HL}] + (1 - s_L)[v_H - P_{HH}]\big] + (1 - b_H)\big[s_L[v_L - P_{LL}] - (1 - s_L)P_{LH}\big] \geq 0
\end{aligned}
$$

Ex-Ante Participation $\Longrightarrow$ Ex-Post Efficiency

The (expected) transfer from the buyer to the seller is

$$\bar{P} = s_L[(1 - b_H)P_{LL} + b_H P_{HL}] + (1 - s_L)[(1 - b_H)P_{LH} + b_H P_{HH}]$$

which implies that the two (IR) can be written as

$$b_H v_H + (1 - b_H)v_L \geq \bar{P} \geq s_L c_L + (1 - s_L)c_H$$

$\bar{P}$ **can be adjusted without affecting (IC)**

Ex-Ante Participation $\Longrightarrow$ Ex-Post Efficiency (Cont'd)

The incentive constraints can be rewritten as

$$(1-b_H)c_H \geq (1-b_H)[P_{LL}-P_{LH}] + b_H[P_{HL}-P_{HH}] \geq (1-b_H)c_L$$

and

$$(1-s_L)v_H \geq s_L[P_{HL}-P_{LL}] + (1-s_L)[P_{HH}-P_{LH}] \geq (1-s_L)v_L$$

Ex-Ante Participation $\Longrightarrow$ Ex-Post Efficiency (Cont'd)

The constraints can be satisfied recursively.

1. Take any vector $(P_{HH}, P_{LH}, P_{HL}, P_{LL})$ that satisfies the Seller's (IC). Then any change in (P_{LL}, P_{LH}) that does not affect $(P_{LL} - P_{LH})$, does not affect the (IC) of S either $\Longrightarrow$ a degree of freedom to satisfy the (IC) of B

2. Any change in $(P_{HH}, P_{LH}, P_{HL}, P_{LL})$ that modifies all payments by the same amount leaves all (IC) unaffected $\Longrightarrow$ a degree of freedom to satisfy the (IR) of both parties

The result extends to arbitrary distributions of types and to quasi-linear environments (d'Aspremont-Gerard Varet (1979))

Interim Participation

The Seller:

$$
\begin{array}{ll}
(IR)_L & b_H P_{HL} + (1 - b_H) P_{LL} - c_L \geq 0 \\
(IR)_H & b_H [P_{HH} - c_H] + (1 - b_H) P_{LH} \geq 0
\end{array}
$$

The Buyer:

$$
\begin{array}{ll}
(IR)_L & s_L [v_L - P_{LL}] - (1 - s_L) P_{LH} \geq 0 \\
(IR)_H & v_H - s_L P_{HL} - (1 - s_L) P_{HL} \geq 0
\end{array}
$$

Interim Participation (Cont'd)

If $m \to 1$ then (IC) of S implies that

$$P_{HH} \approx P_{HL} \equiv \hat{P}$$

and her (IR)

$$\begin{aligned} (IR)_L \quad & \hat{P} - c_L \geq 0 \\ (IR)_H \quad & \hat{P} - c_H \geq 0 \end{aligned}$$

In addition, $(IC)_H$ of B becomes

$$v_H - \hat{P} \geq s_L[v_H - P_{LL}] - (1 - s_L)P_{LH}$$

Interim Participation (Cont'd)

The last inequality implies that the B's (IR) become

$$
\begin{array}{ll}
(IR)_L & \hat{P} \leq v_H \\
(IR)_H & \hat{P} \leq s_L v_L + (1 - s_L) v_H
\end{array}
$$

Collecting all (IR) yields

$$s_L v_L + (1 - s_L) v_H \geq c_H$$

which <u>does not hold</u> if $r \to 1$

Myerson-Sattertwhite (1983)

Theorem (Myerson-Sattertwhite (1983))

There is no allocation which is ex-post efficient, implementable in a Bayesian Nash Equilibrium, and which gives every type nonnegative gains from participation

In the example: type $\theta^B = H$ and $\theta^S = L$ must get informational rents, but the sum of the rents may exceed social surplus in some circumstances:

Interim-(IR) does not allow to freely transfer rents between agents

An Equivalence Result (1)

A standard quasi-linear environment with one principal and one agent who is privately informed with $\theta \in \Theta$

$$U = \theta Q - T$$

$$V = T - c(Q), \qquad c', c'' > 0$$

We consider two alternative approaches to characterization

An Equivalence Result (2)

a) Revelation Principle

1. Restrict attention to direct mechanisms $\Theta \to \mathbb{R}_+^2$ which are (IC)
2. The principal looks for the optimal $(Q(.), T(.))$ satisfying the constraint

b) Constructing a Tariff

1. The Principals post a menu, i.e. a price-quantity tariff
2. The Agent picks one quantity in the tariff given her type

An Equivalence Result (3)

Proposition (Taxation Principle: Rochet (1985))

The two approaches implement the same set of equilibrium allocations

Proof:

Revelation Principle $\Longrightarrow$ Tariff: If $(Q(.), T(.))$ is (IC), then consider the tariff

- $T(Q) = T(\theta) \qquad$ if $Q = Q(\theta)$
- $\infty \qquad$ otherwise

$\blacksquare$

An Equivalence Result (4)

Tariff $\Longrightarrow$ Revelation Principle: Fix a tariff $T(.)$ and let

$$Q(\theta) = \underset{Q}{Max} \quad \theta Q - T(Q)$$

Type θ finds it optimal to report

$$\tilde{\theta} = \underset{\theta'}{argmax} \quad \theta Q(\theta') - T(Q(\theta')) = \theta$$

$\blacksquare$